

ER-Requirements Specification

A1: BountyBust

This project aims to develop a collaborative Q&A platform, designed to reimagine the education world in a compact and fun way for everyone.

The BountyBust website allows you to share your head-scratching homework, problems, and other doubts that you might have in all kinds of areas of interest, presented as tags. This will be posted as bounties, which any hunter can pursue. Whether you're a student or a fully employed person, in your own needs, you can play both roles: post your own bounties, solve others' challenges, or explore and comment on solutions already available.

The platform follows a privilege-based hierarchy that offers various levels of accessibility and functionalities. At the top of the pyramid, administrators are responsible for maintaining order in the system through account regulation and tag management. Following them, we have users whose access to features varies depending on the situation.

When you enter the BountyBust website, you can explore bounties and answers, and using our filtering and search tools, find what you are looking for effortlessly. To become an active participant, however, you'll need to register and customize your profile. Doing so unlocks the full experience, allowing you to create your own bounties, pursue those posted by others, follow specific tags and bounties, and view everything in your own personalized space. Finally, one can find moderators, who are individuals tasked with handling inadequate or misplaced content by deleting it or simply editing a tag.

A point system is in place to reward you for successfully completing bounties, with points granted when the author marks your solution as correct and for each upvote you receive. Based on your points, you can earn a spot on the leaderboard and have your profile highlighted amongst other users.

Additional features include a notification system to track interactions on your content, a responsive website design compatible with all devices, and a section that provides a deeper look into BountyBust and its creators.

A2: Actors and User Stories

A component that outlines the system's actors and their functionalities through user stories, which are a key part of requirements engineering.

1. Actors

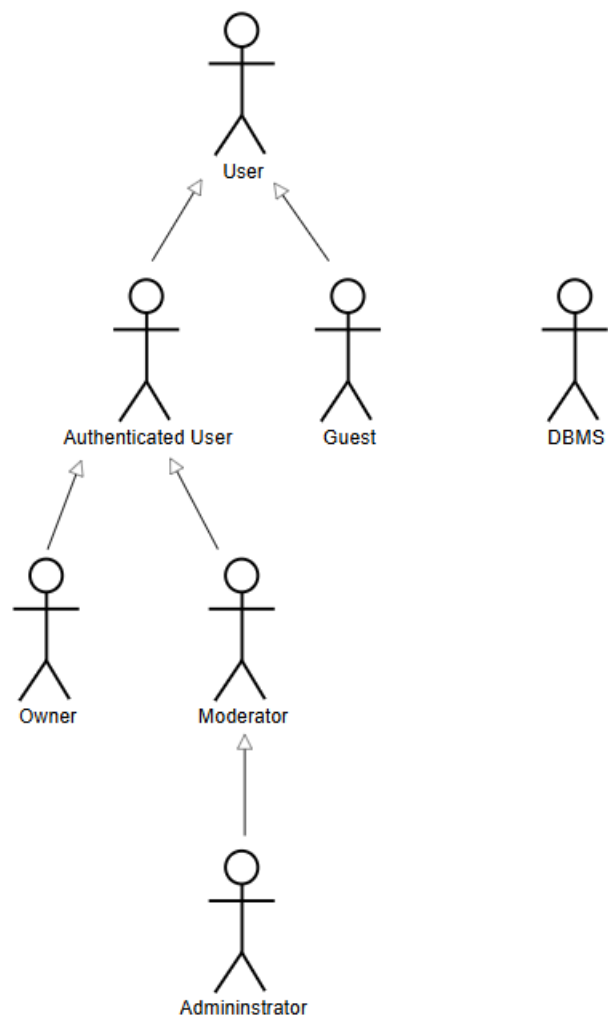


Figure 1: BountyBust Actors.

Identifier	Description
User	A common user with access to standard functionalities like bounties and answers browsing.
Guest	A user who hasn't registered or signed in to the platform in the current session.
Authenticated User	Is allowed to create and respond to bounties; have them displayed in a personal area, as well as interact with others' content.
Owner	Corresponds to the authenticated user responsible for the creation of an item (question, answer, comment) who can edit or delete it.
Moderator	An authenticated user appointed to manage unsuitable content by its deletion or, in case of a simple bounty tag misplacement, its edition.
Administrator	An authenticated user whose main role is to envision and maintain the good welfare of the system through user moderative actions and resource management.
DBMS (Database Managing System)	A system that enables storing, retrieving, editing, and deleting data within the BountyBust ecosystem.

Table 1: BountyBust actors description.

2. User Stories

User				
Identifier	Name	Priority	Description	Responsible Group Member
US01	Homepage	Prototype	As a User, I want to be able to access the home page, so that I can see the featured bounties and tags.	Vitor Pereira
US02	Over Text Search	Prototype	As a User, I want to have a text search correspondence bar, so that I can find the questions I am looking for.	José Archer

US03	See Featured Questions	Prototype	As a User, I want to view the top questions so that, I know which ones have received more engagement.	Vitor Pereira
US04	Browse Through Questions	Prototype	As a User, I want to go through the questions so that, I see the ones I like the most.	Marcelo Lima
US05	Search Across Questions and Answers	Prototype	As a User, I want to broaden my search across questions and answers, so that I can find any related content to my needs.	José Archer
US04	Assistance Features	Product	As a User, I want to be enlightened in unknown or failure contexts, so that I have a better website experience.	Marcelo Lima
US05	Main Features Showcase	Product	As a User, I want to understand the main features of the website, so that I am better informed about what it offers.	Marcelo Lima
US06	About Us Section	Product	As a User, I want an About Us section, so that I can learn more about the BountyBust website's background.	Vitor Pereira
US07	Contacts	Product	As a User, I want to contact the team behind the BountyBust website, so that I can clarify some uncertainties.	Vitor Pereira
US08	Search based on attributes	Product	As a User, I want to search over specific or multiple attributes, so that I can further restrain from what I am seeking.	José Archer
US09	View Recent Questions	Product	As a User, I want to view the most recent questions, so that I can be among the first people to solve them.	Vitor Pereira

US10	View Question Details	Product	As a User, I want to view a question's details, so that I can further inspect it.	José Archer
US11	View User Profiles	Product	As a User, I want to see others' profiles, so that I can learn more about them.	Vitor Pereira
US12	Order search results	Innovation	As a User, I want to order my search results so that the most relevant items are easier for me to find.	José Archer

Table 2: User user stories.

Guest				
Identifier	Name	Priority	Description	Responsible Group Member
US13	Login (DTBMS)	Prototype	As a Guest, I want to log into the system using my website credentials, so that I can access all my account-related information.	José Archer
US14	Register (DTBMS)	Prototype	As a Guest, I want to register myself on the BountyBust website, so that I can access it whenever I want to get onto the system.	José Archer
US15	Logout	Prototype	As a Guest, I want to logout of my account, so that no one without my credentials can access my account on that device.	Marcelo Lima
US16	Recover Password (DTBMS)	Product	As a Guest, I want to be able to recover my password so that I don't worry about forgetting it when logging in.	Marcelo Lima

Table 3: Guest user stories.

Authenticated User				
Identifier	Name	Priority	Description	Responsible Group Member
US17	Personal Feed	Prototype	As an Authenticated User, I want to be able to access a personal feed area, so that I can see content based on my preferences.	Vitor Pereira
US18	Post Questions	Prototype	As an Authenticated User, I want to be able to post questions, so that other people can help me solve my problems.	Vitor Pereira
US19	Answer Questions	Prototype	As an Authenticated User, I want to be able to answer other people's questions, so that I can share my expertise with others.	Marcelo Lima
US20	View my Activity	Prototype	As an Authenticated User, I want to be able to view my activity, so that I can see my posted questions and answers.	Marcelo Lima
US21	View Profile	Prototype	As an Authenticated User, I want to view my profile, so that I have a unique area where my personal information is displayed.	José Archer
US22	Edit Profile	Prototype	As an Authenticated User, I want to edit my profile so that I can keep my personal information up to date.	José Archer

US 23	View Notification List	Product	As a User, I want to have access to a notification list, so that I am presented with platform-relevant content all in one place.	Marcelo Lima
US24	Mark Notification as Read	Product	As a User, I want to mark a notification as read so that I can stay organized.	Marcelo Lima
US25	View Personal Notifications	Product	As an Authenticated User, I want to view my personal notifications, so that I can stay informed about data related to me.	Marcelo Lima
US26	Profile Picture	Product	As an Authenticated User, I want to upload or update a profile picture, so that I can have an extra way of representing myself.	José Archer
US27	Voting System	Product	As an Authenticated User, I want to be able to vote on people's questions and answers, so that I can share my feedback.	José Archer
US28	Comment System	Product	As an Authenticated User, I want to be able to comment on people's questions and answers, so that I can give my opinion.	José Archer
US29	Follow Questions	Product	As an Authenticated User, I want to follow questions, so that I can see their responses and activity over time.	José Archer
US30	Follow Tags	Product	As an Authenticated User, I want to follow a specific tag, so that I	Marcelo Lima

			can find only topics of my interest.	
US31	Point system	Innovation	As an Authenticated User, I want there to exist a point system, so that I can be rewarded for my actions on the website.	José Archer
US32	Leaderboard	Innovation	As an Authenticated User, I want a leaderboard to be available, so that the most skilled problem solvers, with the highest points, can have their profiles showcased.	José Archer
US33	Spend Points in Exclusive Profile Customization	Innovation	As an Authenticated User, I want to be able to spend points in exclusive profile customization, so that I can show it off to others.	José Archer
US34	Achievement/Badge System	Innovation	As an Authenticated User, I want to have an achievement system, so that I can have them displayed in my profile through a unique badge.	José Archer

Table 4: Authenticated User user stories.

Owner				
Identifier	Name	Priority	Description	Responsible Group Member
US35	Content Edition	Prototype	As an Owner, I want to change the content of my questions or answers, so that I can update them if needed.	Marcelo Lima
US36	Content Deletion	Prototype	As an Owner, I want to be able to delete my questions or answers, so that they are no longer available for other users.	Marcelo Lima
US37	Comment Manipulation	Product	As an Owner, I want to delete or edit my comments, so that they no longer have the impact I envisioned previously.	José Archer
US38	Mark Answer as correct	Product	As an Owner, I want to mark someone's answer to my question as correct, so that they are rewarded for their efforts.	Marcelo Lima
US39	Edit Questions Tag	Product	As an Owner, I want to edit my questions' tags so that I can correct any mistakes if they are misplaced.	Marcelo Lima
US40	Notification in reply to Question	Product	As an Owner, I want to receive a notification when someone replies to my question, so that I am warned and quickly prompted to check it out.	Marcelo Lima

US41	Notification on content voting	Product	As an Owner, I want to be notified whenever someone votes on my content, so that I can keep track of others' feedback.	Marcelo Lima
US42	Question Point Reward	Innovation	As an Owner, I want the ability to set a predefined point reward on a question, so that those who answer correctly can be rewarded.	José Archer
US43	Vote on Answer Point Reward	Innovation	As an Owner, I want to be rewarded each time my answer is positively voted on, so that I can have extra points.	José Archer

Table 5: Owner user stories.

Moderator				
Identifier	Name	Priority	Description	Responsible Group Member
US44	Content Moderation	Product	As a Moderator, I want to be able to delete unsuitable content, so that the website is a safer space for everyone.	Vitor Pereira
US45	Question Tag Editing	Product	As a Moderator, I want to be able to edit question tags, so that they more accurately identify the question.	Vitor Pereira

US46	Content Appeal	Innovation	As a Moderator, I want to be able to accept appeals for content reports, so that questions or answers incorrectly marked as unsuitable can be resubmitted.	Vitor Pereira
US47	Manipulate User Points	Innovation	As a Moderator, I want to manipulate user points so that people who earned them unfairly have them removed.	Vitor Pereira

Table 6: Moderator user stories.

Administrator				
Identifier	Name	Priority	Description	Responsible Group Member
US48	Account Moderation	Product	As an Administrator, I want to be able to ban or delete a user's account so they can't post more inappropriate questions or answers.	Vitor Pereira
US49	User Control	Product	As an Administrator, I want to search, view, edit, and create user accounts, so that I can assist in resolving account-related issues.	Vitor Pereira

US50	Support Administrator Accounts	Product	As an Administrator, I want to elect and interact with new administrators so that they can assist with system regulation.	Vitor Pereira
US51	Moderator Management	Product	As an Administrator, I want to grant or revoke privileges from Users so they can become platform moderators.	Marcelo Lima
US52	Adding/Removing Question Tags	Product	As an Administrator, I want to be able to manage question tags, so that the system is kept up to date	Marcelo Lima
US53	Account Appeal	Innovation	As an Administrator, I want to be able to accept appeals for account reports, so that users who have been incorrectly reported can get back access to their blocked accounts	Vitor Pereira

Table 7: Administrator user stories.

3. Supplementary Requirements

Business Rules		
Identifier	Name	Description
BR01	Administrator Behaviour	Administrators are also active participants in the community, with the ability, for instance, to post or vote on questions and answers.
BR02	Edition Indicator	Any question and answer that has been edited after being posted should have a clear notice of those edits.

BR03	User Badges Conditions	User badges should be granted based on the likes and dislikes voted on questions and answers, as well as actions taken by the user (e.g., first question, first answer).
BR04	User Point Attribution	Points are deducted or increased based on the votes attributed to a user's questions or answers. In addition, you receive points when your answer is marked as correct.
BR05	Voting on Own Content	You can't vote on your own questions or answers; that feature will be disabled for you.
BR06	Account Deletion	When your account is deleted, all your content is preserved; it is instead showcased as from an anonymous user.
BR07	Date Consistency	Any questions or answers with time discrepancies can be deleted by moderators.
BR08	Question Reward	All questions will have a standard point reward value. Users who want their question to stand out and be more valuable can deduct from their own points to catch more attention.

Table 8: BountyBust
Business Rules

Technical Requirements

Identifier	Name	Description
TR01	Usability	The BountyBust website should be easy to navigate and meet the user's expectations. All features should be accessible through a simple and fun interface. This is a key factor in how the user perceives their experience, and fundamentally, the effort they make in accessing what they want cannot be a problem.
TR02	Adaptability	The system should behave and change according to your device's display size, without losing functionality. Your

		experience should also be independent of the browser and the machine on which it is being run. This assures we can get to the most users as possible, and the current ones can access the website and their accounts from multiple ways.
TR03	Performance	The system should have response times shorter than 2 seconds to ensure the user's attention. Fluidity across BountyBust supports and complements everything else, as it is a precondition in all digital platforms nowadays.

Table 9: BountyBust
Technical Requirements

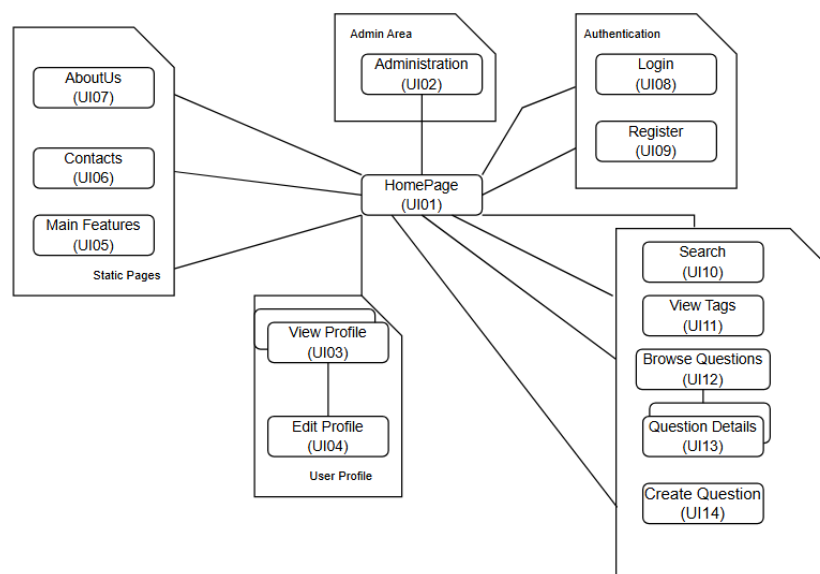
Restrictions		
Identifier	Name	Description
C01	Negative Points	A user cannot have negative points, with 0 being the lowest they can have.
C02	Unique Usernames	Two users on the platform cannot have the same username. This means upon registering, you must choose one who isn't already present in the BountyBust database.

Table 10: BountyBust
Restrictions

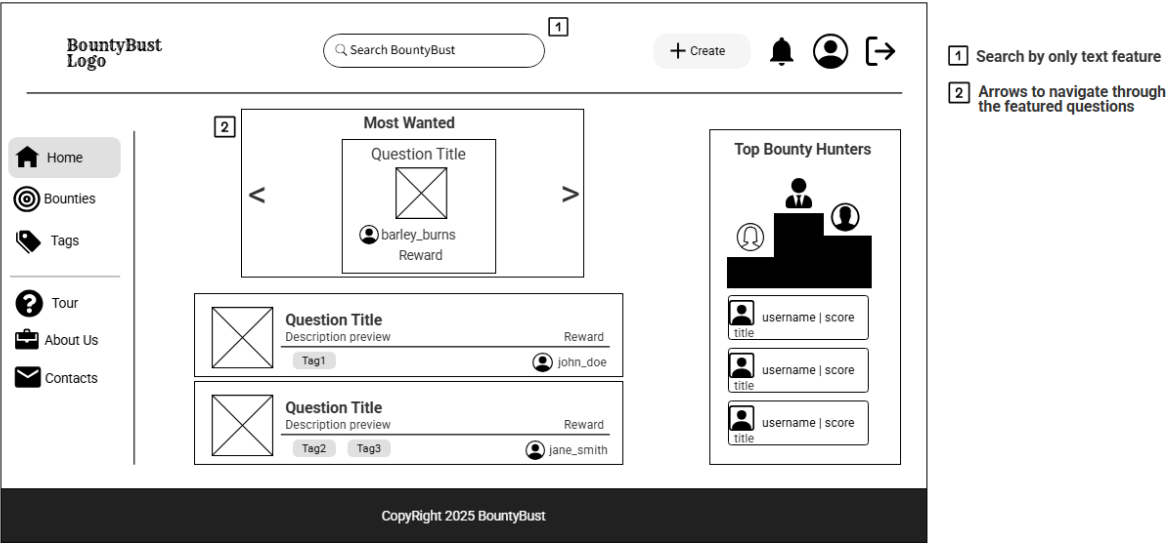
A3: Information Architecture

An artifact that focuses on how the different spaces within the platform are organized and how they relate to each other.

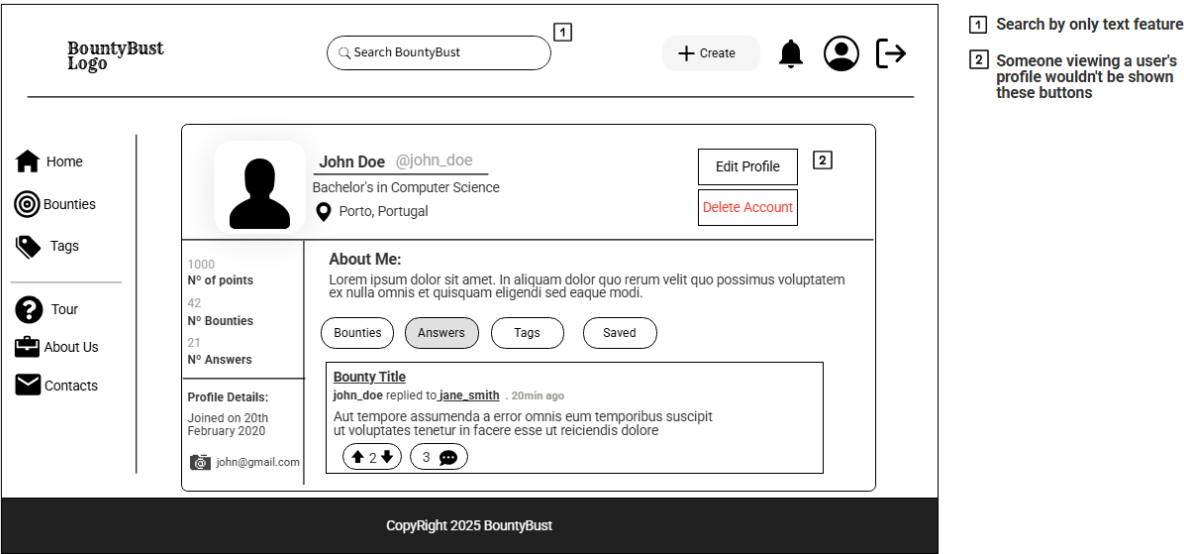
1. Sitemap



2. Wireframes



Homepage Wireframe of the
BountyBust Website



User's Profile Wireframe of the
BountyBust Website

23/09/25

Revision History

No follow-up changes have been made to this document at this time.

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